

Exercise 1

Defining dynamic treatment strategies

Learning Objectives

By the end of this Exercise, be able to map a pharmacoepidemiologic research question to the protocol of a target trial, and understand clinical implications of design choices.

Overview

Questions will be based on the paper Glucagon-like peptide-1 receptor agonists and risk for cardiovascular events in older adults treated with levothyroxine: a target trial emulation, by Du et al. 2025. Reading this paper is not required for participation in this exercise, but you may choose to do so for additional background context. A brief description has been included for your convenience.

Background: Initiation of Glucagon-Like Peptide-1 receptor agonists (GLP-1RA) in patients with type 2 diabetes (T2D) treated with levothyroxine may decrease the required levothyroxine dose due to weight loss or enhance levothyroxine absorption through delayed gastric emptying. These changes may cause thyroid hormone over-replacement and increased risk of atrial fibrillation/flutter (AF/Aflutter) and stroke. This study aims to investigate the impact of GLP-1RA compared to SGLT2 inhibitors on risks of AF/Aflutter and stroke in patients with T2D treated with levothyroxine.

Methods: We used weighted Cox proportional hazards models to estimate the hazard ratio (HR) and 95% confidence interval (CI) for primary and secondary outcomes. Inverse probability weights (IPW) were estimated by pooled logistic regression to account for informative censoring due to treatment discontinuation, treatment initiation, bariatric surgery, initiation of medications suggestive of thyroid disease or interfere with thyroid hormone levels, new diagnosis of a thyroid neoplasm, thyroid cancer, hyperthyroidism, thyroiditis, or other thyroid diseases.

Results: After matching, the study cohort included 2,384 participants in both GLP-1RA and SGLT2 groups. The median follow-up time was 1.05 years. Compared to SGLT2 inhibitors, initiation of GLP-1RA was associated with higher risk of AF/Aflutter (HR: 1.46;

95% CI: 1.28–1.67), while no statistically significant difference was observed between the two groups for stroke (HR: 1.17; 95% CI: 0.98–1.39).

Part 1

Below find the treatment, outcome, and follow-up aspects from the protocol of the target trial Du et al. 2025 aimed to emulate using Medicare data.

	Specification	Emulation
Eligibility	• Age ≥ 65 • Diagnosed with T2D • Stable levothyroxine dose for ≥ 6 months • Willingness to participate and provide informed consent	• Same as the target trial • The base-cohort entry date was defined as the date of patients' first inpatient diagnosis or the second outpatient diagnosis of T2D, identified by relevant ICD-9/10-CM diagnosis codes • Patients entered the study-cohort at the date when they initiate GLP-1RA or SGLT2 inhibitors
Treatments	• Eligible patients were randomly assigned to initiate GLP-1RA or SGLT2 inhibitors in a 1:1 ratio	• We emulated randomization using PS matching with a 1:1 ratio to initiate GLP-1RA or SGLT2 inhibitors
Follow-up	• Patients were followed from the date of randomization until the first occurrence of any of the following events: a study outcome, bariatric surgery, initiation of medications suggestive of thyroid disease or interfere with thyroid hormone levels (e.g., PPIs, prednisone, or prednisolone), new diagnosis of a thyroid neoplasm, thyroid cancer, hyperthyroidism, thyroiditis, or other thyroid diseases (excluding hypothyroidism), loss to follow-up, discontinuation of assigned treatment (GLP-1 or SGLT2), switch to the other studied treatment, death, or the end of the study period (December 31, 2020)	• Same as the target trial • Loss to follow-up was defined by not continuous enrolling in the Medicare Part A/B/D plan

Based on the study protocol, answer the following questions:

- a) What is the target population? Do the eligibility criteria ensure that all patients could plausibly receive either treatment strategy?
- b) Recall that for certain estimation procedures, like inverse probability weighting, individuals may be censored when they deviate from the assigned strategy. With this in mind, is the study estimating the effect of GLP-1RA vs. SGLT2 inhibitors alone, or is it estimating something more specific?

Part 2

Consider the following proposed modifications to the study design, which change one or more components of the original protocol. For each scenario, answer the following questions:

- Does the target population change? Do the eligibility criteria still ensure that all patients could plausibly receive either treatment strategy?
- What do the changes to censoring imply about the treatment strategy now being estimated?
- What biases would you be concerned about introducing by this modification? How could you address them?
- Would these treatment strategies be feasible to implement in a clinical setting?

Scenario 1: Your colleague proposes modifying the follow-up component of the protocol to focus exclusively on treatment adherence and continuation, such that follow-up would end either if they changed treatment, experienced a study outcome (AF/Aflutter, stroke), were lost to follow-up, or the study period ended. Under the modified approach, patients would be censored when they: (1) discontinue their assigned GLP-1RA or SGLT2 inhibitor, (2) switch to the comparison medication, (3) add the comparison medication during follow-up, or (4) discontinue levothyroxine. They argue this approach maintains focus on the true per-protocol effect.

Scenario 2: Your colleague has linked the Medicare claims data to electronic health record data that includes TSH measurements. They propose redefining the eligibility criteria such that TSH is measured at baseline, and the treatment strategies to require continued levothyroxine use with TSH monitoring to adjust levothyroxine if TSH is suppressed in both treatment strategies. Patients who do not have a TSH measurement within 18 months of their most recent test would be censored, since their adherence to the monitoring strategy could no longer be verified. They argue this approach better reflects a clinically actionable strategy by incorporating guideline-recommended thyroid monitoring into the protocol definition.

Scenario 3: Your colleague suggests the SGLT2 inhibitors is less relevant and wants to focus on GLP-1RA alone. They propose redefining the eligibility criteria such that the requirement for ≥ 6 months of levothyroxine is removed, and the treatment strategies to compare initiation of GLP-1RA within 3 months of the first levothyroxine prescription versus initiation of GLP-1RA within 6 months of the first levothyroxine prescription. They argue this approach addresses a more clinically relevant question about the optimal timing of GLP-1RA initiation relative to hypothyroidism treatment.

Hint: Would the treatment strategies be indistinguishable when eligibility criteria are met?

Summary

In this Exercise, we learned how to map elements of a target trial to observed data, and understand implications of changing different protocol components.